Zeigen: Inhalt: Mathematik II (V

ilias.uni-freiburg.de/ilias.php?baseClass=ilrepositorygui&ref_id=4174927

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Übersicht

Meine Kurse und Gruppen

Magazin

KI Tools

Support

Kommunikation

Mehr

Magazin > Lehrveranstaltungen aus HISinOne > Sommersemester 2026 > Mathematisches Institut > Mathematik II (Vorlesung) | SoSe 26

Mathematik II (Vorlesung) | SoSe 26

Aktionen

Inhalt | Info | Einstellungen | Mitglieder | Lernfortschritt | Metadaten | Export | Chat

Zeigen | Verwalten | Sortieren

Neues Objekt hinzufügen | Voransicht als Mitglied aktivieren | Seite bearbeiten

1. From the home page, scroll down...

Mathematik II für Studierende der Naturwissenschaften

Bei Fragen oder Problemen schreiben Sie bitte an: **mfnw@math.uni-freiburg.de**

Aktuelles

Zeigen: Inhalt: Mathematik II (V

ilias.uni-freiburg.de/ilias.php?baseClass=ilrepositorygui&ref_id=4174927

Scuola

Blatt00.ipynb
57,1 kB • Fine

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ak

Magazin > Lehrveranstaltungen aus HISinOne > Sommersemester 2026 > Mathematisches Institut > Mathematik II (Vorlesung) | SoSe 26

Übersicht

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Magazin

KI Tools

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Kommunikation

Mehr

Informationen zu Vorlesung und Klausur

Hier finden sich Infos zur Vorlesung und zur Klausur. Bitte unbedingt lesen!

You can place it where you wish.

Übungsblätter und Lösungshinweise

Abgabe der Übungsblätter

...untill you find Python-Kurs

Python-Kurs

Blatt00

ipynb 58.45 KB Heute, 13:46

2. Click on the file to download it.

Zusätzliche Übungsangebote, kosmic

Forum

Literatur

https://ilias.uni-freiburg.de/ilias.php?baseClass=ilrepositorygui&cmdNode=10v:pc&cmdClass=ilObjFileGUI&cmd=sendfile&ref_id=4205765&origin=dropzone

Zeigen: Inhalt: Mathematik II (V | Project Jupyter | Home

jupyter.org

jupyter

Try Install Community AI Documentation News Social Governance Security About

3. Open a new tab and type 'jupyter.org' on the adress bar to see the current webpage.

4. Click on 'Try' and scroll down...

jupyter

Free software, open standards, and web services for interactive computing across all programming languages

Zeigen: Inhalt: Mathematik II (V | Project Jupyter | Try Jupyter

jupyter.org/try

you can find installation instructions here.

⚠️ **Experimental** ⚠️ several of the environments below use the JupyterLite project to provide a self-contained Jupyter environment that runs in your browser. This is experimental technology and may have some bugs, so please be patient and report any unexpected behavior in the JupyterLite repository.

... untill you find JupyterLab

5. Click on it

JupyterLab

The latest web-based interactive development environment

Jupyter Notebook

The original web application for creating and sharing computational documents

Voilà

Share insights by converting notebooks into interactive dashboards

Kernels

https://jupyter.org/try-jupyter/lab/?path=notebooks%2FIntro.ipynb

Zeigen: Inhalt: Mathematik II (V

Intro.ipynb - Try Jupyter!

jupyter.org/try-jupyter/lab/

File Edit View Run Kernel Tabs Settings Help

/ notebooks /

Name	Modified
cpp-smallpt.ipynb	3mo ago
cpp-third-party-li...	3mo ago
cpp.ipynb	3mo ago
Intro.ipynb	3mo ago
Lorenz.ipynb	3mo ago
r.ipynb	3mo ago
sqlite.ipynb	3mo ago

You will get to this tutorial. You can read it if you want but no worries! We will see something similar together.

Intro.ipynb

Markdown

Notebook

Python (Pyodide)

Introduction to the JupyterLab and Jupyter Notebooks

This is a short introduction to two of the flagship tools created by the Jupyter Community.

> ⚠️ **Experimental!** ⚠️ : This is an experimental interface provided by the JupyterLite project. It embeds an entire JupyterLab interface, with many popular packages for scientific computing, in your browser. There may be minor differences in behavior between JupyterLite and the JupyterLab you install locally. You may also encounter some bugs or unexpected behavior. To report any issues, or to get involved with the JupyterLite project, see the JupyterLite repository.

JupyterLab 🧪

JupyterLab is a next-generation web-based user interface for Project Jupyter. It enables you to work with documents and activities such as Jupyter notebooks, text editors, terminals, and custom components in a flexible, integrated, and extensible manner. It is the interface that you're looking at right now.

For an overview of the JupyterLab interface, see the **JupyterLab Welcome Tour** on this page, by going to `Help -> Welcome Tour` and following the prompts.

> **See Also**: For a more in-depth tour of JupyterLab with a full environment that runs in the cloud, see the JupyterLab introduction on Binder.

Jupyter Notebooks 📓

Simple 2 Python (Pyodide) | Idle Mode: Command Ln 1, Col 1 Intro.ipynb 0

Zeigen: Inhalt: Mathematik II (V

Intro.ipynb - Try Jupyter!

jupyter.org/try-jupyter/lab/

File Edit View Run Kernel Tabs Settings Help

/ notebooks /

New Folder

6. Click on this icon to create a new folder!

Name	Modified
Untitled Folder1	now
cpp-smallpt.ipynb	3mo ago
cpp-th[illegible]-party-li...	3mo ago
cpp.ipynb	3mo ago
Intro.ipynb	3mo ago
Lorenz.ipynb	3mo ago
r.ipynb	3mo ago
sqlite.ipynb	3mo ago

Rename it as you wish

Intro.ipynb

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Simple 2 Python (Pyodide) | Idle Mode: Command Ln 1, Col 1 Intro.ipynb 0

Zeigen: Inhalt: Mathematik II (V

Intro.ipynb - Try Jupyter!

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File Edit View Run Kernel Tabs Settings Help

/ notebooks /

Name	Modified
Python-kurs	52s ago
-smallpt.ipynb	
cpp-third-party-li...	
cpp.ipynb	
Intro.ipynb	
Lorenz.ipynb	3mo ago
r.ipynb	3mo ago
sqlite.ipynb	3mo ago

Name: Python-kurs
Size: 0 B
Path: notebooks
Created: 4/17/26, 3:52 PM
Modified: 4/17/26, 3:53 PM
Writable: true

Double click on your folder
to get to your working place.

Intro.ipynb

Markdown

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Simple 1 Python (Pyodide) | Idle Mode: Command Ln 1, Col 1 Intro.ipynb 0

Zeigen: Inhalt: Mathematik II (V

Intro.ipynb - Try Jupyter!

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File Edit View Run Kernel Tabs Settings Help

/ notebooks / Python-I

Upload Files

Name Modified

Click on this icon to upload your files.

Intro.ipynb

Markdown

Notebook

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Jupyter Notebooks 📓

Simple 1 Python (Pyodide) | Idle Mode: Command Ln 1, Col 1 Intro.ipynb 0

Navigate to the foldere where you stored the file and select it.

Click on 'open' to finalize the upload

I'm sorry my machine's language is Italian

Zeigen: Inhalt: Mathematik II (V

Blatt00.ipynb - Try Jupyter!

jupyter.org/try-jupyter/lab/

File Edit View Run Kernel Tabs Settings Help

/ notebooks / Python-kurs /

Name	Modified
Blatt00.ipynb	now

Intro.ipynb | Blatt00.ipynb

Markdown | Notebook | No Kernel

Python: Mathematische Grundlagen

In der Informatik können wir *Daten* als die Objekte definieren, die wir manipulieren möchten, und *Prozeduren* als die Beschreibungen der Regeln zur Manipulation dieser Daten. Die einfachste Informationseinheit, die wir betrachten können, ist eine gegebene Zahl.

Python math ba … jects we want to manipulate, and **Procedures** as the descriptions of the rules for m … n consider is a given number.

[1]:
```
365 # This is a … ment)
```

[1]: 365

Um den Datentyp … en wir die Funktion `type()`:

To check the nature of a datum, we use the primitive function `type(x)`, which prints the data type of the input value `x`.

[2]:
```
type(365)
```

[2]: int

In Python, beschreibt `int` eine Variable des Typs **integer**. Dabei handelt es sich um ganze Zahlen (wie zum Beispiel 5, -10 oder 0), die keine Nachkommastellen besitzen.

In Python, `int` stands for **integer**. An integer is a whole number, positive or negative, without decimals, of unlimited length. It is one of the most fundamental data types in this language.

Select Kernel

Select kernel for: "Blatt00.ipynb"

Python (Pyodide)

☐ Always start the preferred kernel

No Kernel | Select

Finally you have to select the Kernel. The preselected one should be fine !

Just press 'select'

Simple 1 No Kernel | Initializing Mode: Command Ln 1, Col 1 Blatt00.ipynb 0

Zeigen: Inhalt: Mathematik II (V

Blatt00.ipynb - Try Jupyter!

jupyter.org/try-jupyter/lab/

File Edit View Run Kernel Tabs Settings Help

/ notebooks / Python-kurs /

Name	Modified
Blatt00.ipynb	now

Congratulations, you did it.

Intro.ipynb | Blatt00.ipynb

Markdown | Notebook | Python (Pyodide)

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Python math basics In computing, we can define **Data** as the objects we want to manipulate, and **Procedures** as the descriptions of the rules for manipulating those data. The simpler datum we can consider is a given number.

[1]:
```
365 # This is a number (and the rest of the line is a comment)
```

[1]:
```
365
```

Um den Datentyp einer Informationseinheit abzufragen, verwenden wir die Funktion `type()`:

To check the nature of a datum, we use the primitive function `type(x)`, which prints the data type of the input value `x`.

[2]:
```
type(365)
```

[2]:
```
int
```

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Simple | 2 | Python (Pyodide) | Idle | Mode: Command | Ln 1, Col 1 | Blatt00.ipynb | 0